

NAUVAL ZULFIKAR



Mechanistic Interpretability & LLM Security Researcher · Responsible AI candidate Bandung, Indonesia (relocatable to Copenhagen, 2026-09-01) zulfikar.nauval1998@gmail.com · +44 7300 469048 · +62 821 2567 2264

[LinkedIn](#) · [GitHub](#) · [Portfolio](#)

Visa status: UK Graduate Visa (active). Indonesian passport. Eligible for EU Researcher (Hosting Agreement) visa for European PhDs.

PROFILE

MSc Business Analytics graduate from Aston University with a First Class degree, specialising in NLP and mechanistic interpretability of large language models (LLMs). My MSc dissertation focused on blockchain-integrated LLM analysis for enhancing supply chain information systems, supervised by Dr. Ammar Al-Bazi. I am keen to contribute to research on LLM security and false information mitigation under Prof. Isabelle Augenstein at the University of Copenhagen.

EDUCATION

Aston University, Birmingham, UK

MSc Business Analytics, First Class (1:1) · EQF Level 7 · Sep 2023 – Sep 2024

- **1st Winner, Aston Enterprise Scholarship 2023** (full-tuition competitive award)
- **Aston Pitching Contest 2024** runner-up (AI-powered supply chain proposal)
- **Distinctions:** Decision Models, Effective Management Consultancy, Software Analytics
- **Dissertation:** *Enhancing Supply Chain Information Sharing Systems Through Blockchain: Integrated Customer Reviews LLM Analysis: A Smart Contract Approach* (supervised by **Dr. Ammar Al-Bazi**)

Ritsumeikan Asia Pacific University, Beppu, Japan

BBA Marketing · EQF Level 6 · Oct 2016 – Sep 2019

- APU 50% Tuition Reduction Scholarship + JASSO Scholarship
- **Distinctions:** Business Data Analysis, Consumer Behaviour, Marketing Research

Indonesia AI

NLP Engineer Training · EQF Level 5 · May – Nov 2023

- Contributed to the *2024 Indonesia Election Aspect-Based Sentiment Analyser* project
- **Distinctions:** Python Programming, Text Summarisation, Voice Recognition

Qarir Generator (Dubai, UAE)

Data Science Global Accelerator · EQF Level 5 · Apr – Jun 2023

- Led sentiment-analysis development for the *Pfizer Public Sentiment Analysis* project
- **Distinctions:** Python, SQL, Pandas/NumPy

RESEARCH-RELEVANT PROJECTS

LLM-Generated Adaptive Shipper Decision Rules

2026

- Developed an end-to-end RAG system using gpt-4o-mini, augmented with retrieval over a curated behavioural-literature knowledge base. Fully-traceable rule-based fallback.
- github.com/nauvalZulfikar/LLM-Shipper-Profiles

CV-JD Suitability Checker

2024-11

- Implemented a transformer-based semantic matching system using DeBERTa-v3, outperforming lexical ATS baselines. Full ML pipeline: training, evaluation, serving.

MSc Dissertation Pipeline (DeBERTa-v3 fine-tune)

2024

- Fine-tuned DeBERTa-v3 on multi-tier supplier review data for multi-aspect classification. Evaluated emergent biases against literature baselines. Embedded in a larger decision system.

Crowd-Shipping ABM (450-run behavioural parameter sweep)

2026

- Conducted a controlled experiment over reward and supply-density grids; literature-anchored results. Acceptance 45.34% (n=20) to 79.80% (n=100).

VRPOD MILP Optimisation

2026

- Replicated Archetti et al. (2016) VRPOD with `scipy.optimize.milp`, achieving a 38.5% cost reduction on a Birmingham-inspired toy instance.

INDUSTRY EXPERIENCE

Public Works & Spatial Planning Department (PUTR), Bandung

Data Scientist & Project Manager · Sep 2022 – Present

- Automated ETL pipelines and developed self-service dashboards, achieving a 45% faster project delivery and 20% data-efficiency gain.

Syncwell, Birmingham, UK

Marketing Data Scientist · Jul – Sep 2024

- Implemented predictive ML and A/B testing strategies, resulting in a 40% CTR increase and 22% follower growth.

PCOS Challenge, Birmingham, UK

Strategic Data Scientist Specialist · Jun – Aug 2024

- Developed a customer attribution model and automated report text extraction, reducing manual workload by 80%.

Bank Muamalat Indonesia, Jakarta

Lead Data Scientist (Leadership Development Programme) · Nov 2019 – Aug 2022

- Developed NLP models for credit portfolio analysis and risk scoring, achieving a 5.3% investment return.

PUBLICATIONS

Zulfikar, N. (2025) *Performance Analysis in Sport Footwear Sales Prediction Using Machine Learning*. Working paper on comparative evaluation of ML regressors for retail sales forecasting accuracy.

TECHNICAL SKILLS

ML/NLP: PyTorch, HuggingFace Transformers, DeBERTa-v3 fine-tuning, gpt-4o-mini API + RAG, sentence-transformers, classification, regression, time-series forecasting, recommender systems, attribution modelling

Languages/Infrastructure: Python, SQL, R, PySpark, NumPy, Scipy, pandas, scikit-learn, FastAPI, Streamlit

Mathematical/Formal: MILP, Discrete-choice modelling, Agent-Based Modelling, Discrete-Event Simulation, Digital Twin architecture, Bayesian methods, causal inference

Governance: Model validation, model reliability, audit-grade documentation, model staleness monitoring, A/B testing, AWS, Docker, Git, ETL/orchestration, Power BI, Tableau

LANGUAGES

Indonesian (Native) · **English** (Fluent, IELTS 7.0 overall) · **Japanese** (Intermediate)

REFEREES

Dr. Ammar Al-Bazi

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